

Accessible Cyber-Physical Activities

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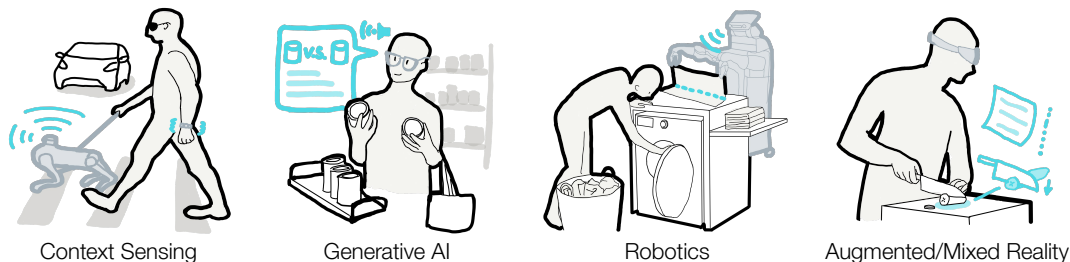


Figure 1: Illustrative scenarios of accessible cyber-physical systems supporting Activities of Daily Living, including Context Sensing, Generative AI, Robotics, and Augmented/Mixed Reality.

Abstract

Recent advances in context sensing, generative AI, robotics, and augmented/mixed reality have created unprecedented opportunities to design cyber-physical systems that meaningfully support accessibility and inclusion in everyday life. These intelligent systems can sense the environment, interpret user goals, and provide adaptive, real-time support, enabling people with disabilities to engage confidently in daily activities ranging from cooking and personal care to creative expression and skill acquisition. However, realizing this potential requires addressing critical challenges in sensing, interaction design, adaptability, and evaluation to ensure systems are trustworthy, user-centered, and extend rather than replace human capabilities. This workshop brings together researchers and designers across HCI, AI, AR/VR, accessibility, and robotics to explore the design and evaluation of accessible cyber-physical systems. For details, please visit <https://accessible-cps.github.io/>.

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CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**.

Keywords

Activities of Daily Living (ADLs), Accessibility, AI, Physical Interaction, Sensing, VR/AR/XR/MR, Robotics, Learning

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1 INTRODUCTION

A powerful convergence of context sensing [4, 11, 13, 26, 31], generative AI [5, 6, 9, 32], robotics [7, 22, 24], and augmented/mixed reality [12, 19, 25, 27, 30] is creating an unprecedented opportunity to design cyber-physical systems that directly enhance accessibility and inclusion in everyday life [14, 16]. These systems can sense the environment, interpret user goals, and deliver adaptive, real-time support—enabling individuals with disabilities to independently engage in everyday activities and other complex physical tasks with confidence.

This shift is especially impactful for accessibility, where technologies like AI-generated instructional support [9, 32] and multimodal interfaces [12, 27] can break down long-standing barriers in domains such as cooking [14, 19, 23], personal care, and creative expression [8, 18], and consuming multimedia [20, 21, 28, 29]. Specifically, these can be valuable from dynamically guiding a user through a multi-step recipe [2, 3, 10, 14, 15, 17], to enabling non-visual skill acquisition [6], to providing intelligent tactile and spatial feedback through robotics [22, 24]. The integration of such cyber-physical systems has the potential to improve the quality of life for people with disabilities by enhancing task efficiency and fostering greater engagement in everyday activities. However, significant design challenges remain: How can we ensure these tools are authorable, adaptable, and trustworthy? How can we design interfaces that extend rather than replace human capability—particularly for users with sensory, motor, or cognitive disabilities? Yet, there have not been many gatherings that bring all these communities together.

This workshop aims to bring together a diverse, interdisciplinary community at UIST—including researchers in HCI, accessibility, AI, AR/VR, and assistive robotics—to chart the future of accessible cyber-physical systems. Through a combination of invited talks, panel discussions, collaborative brainstorming and synthesis, we will explore how these technologies can be leveraged to support not just interaction, but inclusion, independence, and human flourishing. The organizing committee will synthesize insights and ideas generated during the workshop into a report that outlines design directions and opportunities for future research and collaboration.

2 Topics

We will explore the state-of-the-art in accessible cyber-physical systems and foster collaboration among researchers from diverse backgrounds. We aim to address the following key questions:

- How can intelligent systems accurately sense and interpret the physical and social context of everyday activities?
- What interaction paradigms enable real-time, in-situ assistance without disrupting the user's flow or agency?
- What design principles support long-term engagement, trust, and co-adaptation between humans and systems?
- How can we rigorously evaluate these assistive systems in real-world settings?
- What everyday tasks or user groups remain underserved by current accessible cyber-physical systems, and why?

We invite researchers whose work contributes to the future of assistive cyber-physical systems. Example topics include context understanding and multimodal perception; adaptive interfaces; end-user development and authoring; human-centered AI approaches and mixed-initiative interaction strategies [1]; embodied interaction via AR/MR and spatial computing; assistive robotics; accessibility and inclusion in everyday environments; and evaluation methods tailored for in-the-wild deployments.

This workshop is not intended to be a general catch-all. Instead, we seek contributions that critically engage with the design, deployment, and evaluation of accessible cyber-physical systems. Whether grounded in HCI, robotics, AI, AR/VR, or accessibility, we welcome work that is technically rigorous, user-centered, and oriented toward tangible impact.

3 Workshop Plan

Opening Session. The workshop will commence with a brief opening session to introduce the agenda and objectives for the day.

Keynotes (90 min). We plan to invite 2 keynote speakers in the field who will each present for 30 minutes with 15-minute Q&A sessions. They will talk about insights into the latest advancements and future trends in enhancing accessibility in different application scenarios. The keynotes will provide some background information for the following panel and the group discussion.

Panel Discussion (45 min). To prepare for a group discussion on challenges and opportunities of addressing accessibility issues in cyber-physical activities, organizers will invite a panel of 3-5 experts in the fields. During the panel session, participants will be encouraged to engage actively with the panelists. Organizers will moderate this dialogue by collecting questions from participants in advance through Slido. By leveraging the expertise of the panelists and the input from attendees, the session aims to generate valuable discussions and actionable strategies for advancing the field.

Introduction with Poster Session (45 min). The participants will be grouped based on interests, and each group will have diverse backgrounds. The participants will then introduce themselves and engage in icebreaking activities. During the introduction, participants may discuss a range of topics, such as interactions and systems they have constructed relevant to ADLs, the challenges they encountered, or ideas for future research. We plan to host an interactive poster session that fosters in-depth discussions and networking among participants during the introduction session. Posters will be displayed in a dedicated area during the session. Authors will stand by their work to engage with attendees.

Group Discussion and Synthesis (90 min). The participants will be divided into subgroups and engage in a collaborative discussion around the questions outlined in Section 2. Organizers will provide example discussion topics that touch upon critical aspects relevant to making ADLs more accessible. Building on previous keynotes and panel discussions, participants will be encouraged to envision how technical innovations from the UIST community can be integrated into various applications to make the experience more accessible.

In the latter half of the session, all participants will come together to present and synthesize discussion items and results. Each participant will briefly describe one item (e.g., a new sensing technology to facilitate ADLs). Based on these items, participants create an affinity diagram, which will visualize common themes across research areas and disciplines, revealing unique perspectives, and mapping challenges and opportunities. A one-pager summarizing the key findings will be synthesized together as a written output.

Closing Session. The workshop will conclude with a summary of key takeaways and a discussion of the next steps for fostering ongoing collaboration and innovation in the field.

4 Call for Participation

We invite participants and interactive demos from the UIST community who are experienced in or interested in building interfaces for accessible cyber-physical activities. We particularly welcome researchers, designers, and practitioners in interaction techniques, wearables, toolkit development, embedded AI, computational sensing, and personalized modeling.

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